

1- Route Map

Home to uni Route BFS:

[Home]

↓ 0.2 km

[SFC Road Block 4] (R1)

↓ 0.6 km

[A1 Road] (R2)

↓ 1.2 km

[Dygh Road] R3

↓ 1.0 km

[Shahrah-e-faisal] R4

↓ 1.5 km

[Rashid Minhas Rd] R5

↓ 2.4 km

[Johar Mor] R6

↓ 1.5 km

[Nipa chaurangi] R7

↓ 0.5 km

[University Road] R8

↓ 1.2 km

[Fuwast University] R9

$$\text{Route 1: } 0.2 + 0.59 + 1.2 + 1.0 + 1.5 + 2.4 + 1.5 + 0.5 + 1.2 = 10.09$$

$$\text{Route 2: } 0.2 + 0.59 + 1.2 + 1.0 + 1.5 + 2.1 + 1.2 < 1.7$$

2. Graph Representation.

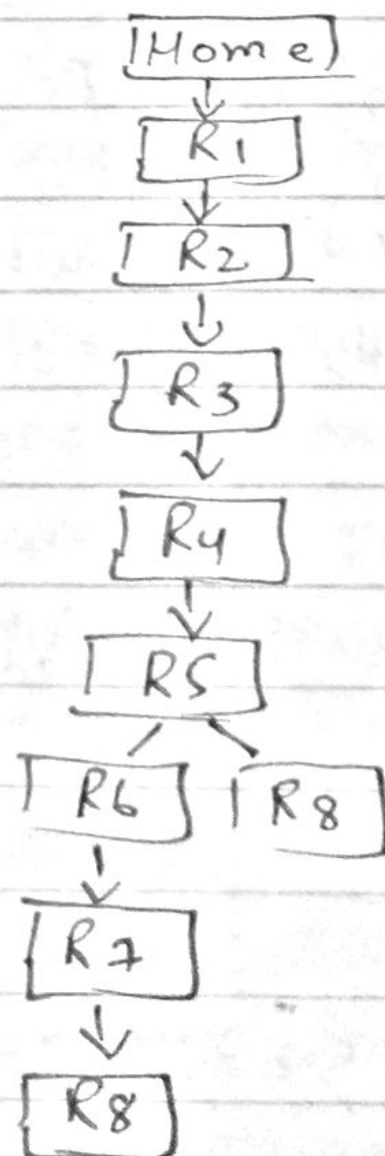
Adjacent Nodes

R_1
Home, R_2
 R_1, R_3
 R_2, R_4
 R_3, R_5
 R_4, R_6, R_8
 R_5, R_7
 R_6, R_8
 R_5 (bypass, R_7, uni)
 R_8

Node

Home Shah faisal block 4
 R_1 (Shah faisal Block 4 Rd)
 R_2 (AI Road)
 R_3 (Deigh Road)
 R_4 (Shahrah-e-faisal)
 R_5 (Rashid Minhas Rd)
 R_6 (Johar Mor)
 R_7 (Nipa chowringi)
 R_8 (University Rd)
uni (Federal urdu uni)

3. BFS Tree:



BFS Path Trace:

- Step 0: Initializing

$$Q = \{ \text{Home} \}$$

$$V = \{ \text{Home} \}$$

Step 1: Dequeue Home $\rightarrow$ Expand Home $\rightarrow$ R1 (SFC Block 4 Rd)

$$Q = \{ R1 \}$$

$$V = \{ \text{Home}, R1, R2 \}$$

Step 2: Dequeue R1 (SFC Block 4 Rd) $\rightarrow$ Expand R1 $\rightarrow$ R2 (A1 Road)

$$Q = \{ R2 \}$$

$$V = \{ \text{Home}, R1, R2 \}$$

Step 3: Dequeue R2 (A1 Road) $\rightarrow$ Expand R2 $\rightarrow$ R3 (Drigh Road)

$$Q = \{ R3 \}$$

$$V = \{ \text{Home}, R1, R2, R3 \}$$

Step 4: Dequeue R3 (Drigh Road) $\rightarrow$ Expand R3 $\rightarrow$ R4 (Shahrah-e-faisal)

$$Q = \{ R4 \}$$

$$V = \{ \text{Home}, R1, R2, R3, R4, R5 \}$$

Step 5: Dequeue R4 (Shahrah-e-faisal) $\rightarrow$ Expand R4 $\rightarrow$ R5 (Rashid Minhas Rd)

$$Q = \{ R5 \}$$

$$V = \{ \text{Home}, R1, R2, R3, R4, R5 \}$$

Step 6: Dequeue R_5 (Rashid Minhas Rd) $\rightarrow$
Expand $R_5 \rightarrow R_6$ (Johar Mor), R_8 (Uni Road)

$$Q = \{R_6, R_8\}$$

$$V = \{Home, R_1, R_2, R_3, R_4, R_5, R_6, R_8\}$$

Step 7: Dequeue R_6 (Johar Mor) $\rightarrow$ Expand $R_6 \rightarrow R_7$
(Nipa Chowrangī)

$$Q = \{R_8, R_7\}$$

$$V = \{Home, R_1, R_2, R_3, R_4, R_5, R_6, R_8, R_7\}$$

Step 8: Dequeue R_8 (University Rd) $\rightarrow$ Expand $R_8 \rightarrow$
Uni (Federal Urdu University)

$$Q = \{R_7, Uni\}$$

$$V = \{Home, R_1, R_2, R_3, R_4, R_5, R_6, R_8, R_7, Uni\}$$

Step 9: Dequeue R_7 (Nipa Chowrangī) $\rightarrow$ Expand R_7
 $\rightarrow R_8$ (University Rd $\rightarrow$ Already visited)

$$Q = \{Uni\}$$

Step 10: Dequeue Uni (Federal Urdu uni)
Goal found!

Shortest Path:

Home $\rightarrow R_1 \rightarrow R_2 \rightarrow R_3 \rightarrow R_4 \rightarrow R_5 \rightarrow R_8$ uni